

Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

```
data = [  
    ['Sunny', 'Warm', 'Yes'],  
    ['Sunny', 'Cold', 'Yes'],  
    ['Rainy', 'Warm', 'No'],  
    ['Sunny', 'Warm', 'Yes']  
]
```

```
hypothesis = ['0', '0']
```

```
for row in data:
```

```
    if row[-1] == 'Yes':
```

```
        for i in range(len(hypothesis)):
```

```
            if hypothesis[i] == '0':
```

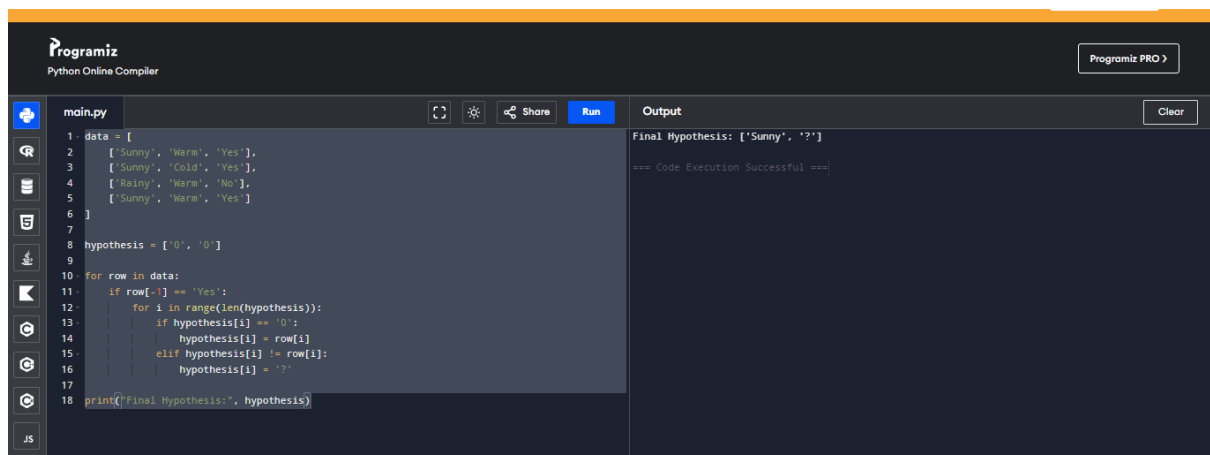
```
                hypothesis[i] = row[i]
```

```
            elif hypothesis[i] != row[i]:
```

```
                hypothesis[i] = '?'
```

```
print("Final Hypothesis:", hypothesis)
```

output



```
Programiz  
Python Online Compiler  
Programiz PRO >  
main.py  
1 data = [  
2     ['Sunny', 'Warm', 'Yes'],  
3     ['Sunny', 'Cold', 'Yes'],  
4     ['Rainy', 'Warm', 'No'],  
5     ['Sunny', 'Warm', 'Yes']  
6 ]  
7  
8 hypothesis = ['0', '0']  
9  
10 for row in data:  
11     if row[-1] == 'Yes':  
12         for i in range(len(hypothesis)):  
13             if hypothesis[i] == '0':  
14                 hypothesis[i] = row[i]  
15             elif hypothesis[i] != row[i]:  
16                 hypothesis[i] = '?'  
17  
18 print("Final Hypothesis:", hypothesis)  
JS  
Output  
Final Hypothesis: ['Sunny', '?']  
=== Code Execution Successful ===  
Clear
```